Confusion, Agency, and Metacognitive Instruction in an Intelligent Tutoring System: An Ability-Moderated Account

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Abstract. This study investigates the interaction between confusion, frustration, and agency under metacognitive instruction in a logic proof tutoring system (N = 249). The tutor teaches two domain-specific metacognitive skills, a Plan/Act/Reflect cycle adapted for logic proof construction and a Target Variable Strategy (TVS) for backward-chaining through logical rules, under varying degrees of learner agency (forced, nudge, optional). Using Control-Value Theory (CVT) as the explanatory framework, we report three findings. First, prior ability moderates which affect responds to metacognitive instruction. High-ability students show reduced frustration and increased confusion, while low-ability students show no affect differences. We call this a frustration-confusion swap: metacognitive instruction reduces task-level struggle while introducing metacognitive-level challenge. Second, nudge prompting generates more confusion than forced prompting in high-ability students. Prior work showed that agency moderates the functional consequences of confusion. Our data extend this by showing that agency also moderates confusion generation. Third, problem-type choice, an independent agency dimension, trends in the same direction. The results suggest that confusion under metacognitive instruction reflects effortful learning of domain-specific metacognitive skills. Furthermore, agency moderates both the generation and functional consequences of confusion.

1 INTRODUCTION

Intelligent tutoring systems (ITSs) increasingly embed metacognitive support alongside domain instruction [2, 24, 6], yet relatively little is known about how students *experience* this support affectively. ITS Research on affect has focused primarily on emotions during domain problem solving [9, 4]. However, even when affective states are measured alongside metacognitive scaffolding [5, 11, 3], prior research has treated affect as related to learning rather than as a response to the metacognitive instruction. Even less is known about how student affective experience varies by prior ability or the degree of agency afforded by the system. This paper investigates the relationship between confusion, agency, and metacognitive instruction in a logic proof tutoring system, using Control-Value Theory (CVT) as the explanatory framework.

1.1 Metacognitive instruction vs. scaffolding

Metacognitive support in ITSs can take two forms. *Scaffolding* provides temporary assistance for skills students are developing, fading as competence grows [28, 19]. *Instruction* teaches new strategies intended for permanent internalization, constituting a genuine learning experience rather than temporary support [26].

The Deep Thought (DT) ITS used in this study teaches two domain-specific metacognitive skills. First, it teaches the Target Variable Strategy (TVS), a backward-chaining strategy in which students identify the target conclusion, determine which variables to derive, and select appropriate logical rules. Although students encountered a version of TVS adapted for probability earlier in the same course [1], the propositional logic-specific TVS requires different variable identification and rule selection strategies, managing forward and backward chaining through 16 logical rules rather than probability calculations. This makes TVS in logic a substantively distinct skill. Second, DT provides instruction in a Plan/Act/Reflect cycle adapted for logic proof construction, in the tradition of [22]. In the Plan phase, students enter goals to derive into a plan stack, specifying parent lines, elimination terms, and rules. The plan interface supports both forward chaining (from available logic statements called premises) and backward chaining (from the target conclusion). During planning, TVS is one backward-chaining strategy students may apply but are not required to. In the Act phase, students populate any remaining parameters for each plan step and execute it by clicking a “Do it” button that applies the planned rule on the proof canvas. Plan and Act may be interleaved step by step rather than completed as a full plan before any execution. In the Reflect phase, an end-of-problem dialog prompts students to type a free-text reflection on how the planning strategy helped them construct the proof, and to report mental effort and help preferences on Likert scales. A post-tutor reflection dialog at the session end assesses perceived mastery of the strategies. Neither Plan/Act/Reflect nor the logic-adapted TVS is a scaffold for an existing skill; both are genuinely new instruction.

The instructional goal is internalization, not fading. Unlike scaffolding that is withdrawn as competence develops, Plan/Act/Reflect and TVS are strategies students are expected to adopt as permanent components of their problem-solving repertoire. The forced, nudge, and optional prompting modes vary how the tutor promotes initial engagement with these strategies. This distinction matters for interpreting affective responses. If metacognitive support were purely scaffolding, confusion might signal that the support is unnecessary, reflecting a mismatch between the student’s existing competence and the imposed structure [13]. But if the support is genuine instruction in new metacognitive skills, confusion may instead signal productive cognitive disequilibrium from acquiring unfamiliar competencies [10, 14, 27]. Domain ability does not guarantee metacognitive proficiency [26]. A student who solves logic problems effectively

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may still have no experience with structured planning or backward-chaining strategies adapted for formal proofs.

1.2 Control-Value Theory

Pekrun's [20] Control-Value Theory (CVT) proposes that learners' appraisals of control and value shape achievement emotions and their consequences for performance, motivation, and well-being. In the original CVT framework, frustration is classified as an activity-related achievement emotion, experienced when a student expects failure and anticipates negative consequences, oriented toward the task itself rather than the learning process. The updated CVT [21] expanded its scope to a general theory of human emotion and introduced a three-dimensional taxonomy. CVT classifies confusion as an epistemic emotion appraised in response to incongruity that could be modeled as:

$$\text{Confusion} \propto \frac{\text{Incongruity} \cdot \text{Value}}{\text{Control}} \tag{1}$$

Confusion increases with the degree of incongruity encountered and the value assigned to the task, and decreases with perceived control. Agency, the degree to which a student has choice over their own behavior, maps onto perceived control in this formulation. Metacognitive support that reduces agency lowers perceived control and should therefore increase confusion when learners encounter incongruity.

1.3 Frustration and confusion as distinct constructs

Frustration and confusion are frequently treated as interchangeable negative affects in tutoring system research, yet they have distinct antecedents, durations, and learning consequences. D'Mello and Graesser [9] demonstrated that confusion arises at productive impasses, moments of cognitive disequilibrium that prompt deeper processing, while frustration signals unresolved impasse and predicts disengagement. D'Mello et al. [10] further showed that confusion predicts learning gains under conditions of productive cognitive engagement, while frustration does not. They also demonstrated that prior knowledge moderates confusion's effects: "it takes a modicum of knowledge to know what one knows and does not know," so low-knowledge learners may fail to detect impasses that would trigger confusion, while high-knowledge learners can both detect and resolve the resulting disequilibrium. Baker et al. [4] extended this analysis, showing that confusion and frustration occupy distinct but related positions on a shared affective continuum with different antecedents and consequences.

CVT specifies the appraisal mechanism that produces confusion (high incongruity and low control), while D'Mello and Graesser specify the functional outcome (deeper processing, if the confusion is resolved). Together, they predict that the *source* of incongruity determines whether confusion is productive: incongruity from encountering genuinely new material (metacognitive instruction) should produce productive confusion, while incongruity from unnecessary constraint (scaffolding mismatch) should not.

1.4 Two sources of incongruity

When students receive metacognitive support alongside domain instruction, the default interpretation of any resulting confusion is *scaffolding mismatch*: students with established problem-solving approaches encounter imposed structure that conflicts with their existing strategies. On this account, the structure itself is the source of incongruity, and the resulting confusion reflects unnecessary cognitive load rather than productive learning [13].

However, when the metacognitive support constitutes genuine instruction in new skills rather than scaffolding of existing ones, an alternative interpretation applies. Plan/Act/Reflect and TVS are domain-specific strategies intended for internalization; neither is a scaffold for a skill students already possess. On this account, incongruity arises from acquiring unfamiliar metacognitive competencies, and confusion reflects *effortful metacognitive learning*, productive cognitive disequilibrium from reconciling new strategies with existing practice.

These sources are not mutually exclusive, but data can be used to distinguish between them. If confusion reflects pure scaffolding mismatch, then imposing metacognitive structure on students who already perform well should increase *both* frustration and confusion (or leave frustration unchanged while increasing confusion). If confusion instead reflects effortful metacognitive learning, we would expect frustration to *decrease* because the instruction reduces task-level struggle, while confusion *increases* from the new metacognitive challenge. We test this prediction in RQ1.

1.5 Agency and the confusion appraisal pathway

Self-determination theory (SDT; [25]) conceptualizes autonomy as a continuum of regulatory styles. The experimental conditions in this study vary student agency along a prompting mode dimension. *Forced* prompting drastically reduces choice over metacognitive behavior, requiring a completed plan on designated training problems before the student can proceed. *Nudge* reduces agency by reminding students to plan while preserving the option to decline. Finally, *Optional* allows planning mode but does not prompt for planning before starting the proof. SDT predicts that forced prompting produces introjected regulation ("I have to do this"), while nudge allows identified regulation ("I see this is useful, and I choose to do it").

The CVT formula predicts that reducing agency (control) should increase confusion. However, agency may also moderate the *functional consequences* of confusion. Droujkov et al. [8] found that in a game-based learning environment, the inverse relationship between average confusion episode duration and intrinsic motivation appeared only under full agency. Under conditions that limited agency, this relationship disappeared. Their finding suggests that agency changes what confusion *does*, i.e., whether it connects to motivational outcomes, rather than simply whether it occurs. This raises the question of whether agency also moderates how confusion is *generated*: under forced prompting, students may comply behaviorally without the depth of cognitive engagement that produces confusion from incongruity.

Schoenfeld [26] demonstrated that expert problem solvers employ a plan-act-reflect cycle that novices lack, and that teaching this cycle explicitly improves mathematical problem-solving. However, the metacognitive control process itself is cognitively demanding, requiring sustained effortful self-regulation that students may experience as confusion even when it is productive. The degree to which students engage with this process, rather than merely performing it under compulsion, should determine whether confusion appears.

1.6 Prior work on confusion, ability, and metacognitive support in learning systems

Several lines of prior research converge on the questions addressed here. Lehman, D'Mello, and Graesser [15] directly tested who benefits from induced confusion, using false feedback to trigger cognitive disequilibrium. Only students with *both* high cognitive ability and high cognitive drive reported more confusion after false feedback. These students also showed transfer learning gains, but only when meaningfully confused. Students with low ability or low drive neither experienced productive confusion nor benefited from it. This finding is the closest precedent for our ability-moderated account: it establishes that confusion is not uniformly experienced and that individual differences determine whether confusion connects to learning.

Lodge and Kennedy [16] proposed a theoretical model of "zones of optimal and sub-optimal confusion," integrating cognitive disequilibrium, desirable difficulties, productive failure, and impasse-driven learning. They argued that confusion is productive only when learners receive appropriate support at the right time and have the capacity to resolve the disequilibrium. Our two-source incongruity framework (Section 1.4) aligns with their model: scaffolding mismatch places learners in a zone of sub-optimal confusion, while effortful metacognitive learning places them in the optimal zone.

Richey et al. [23] examined confusion and frustration jointly, measured as "confrustion," in an erroneous examples tutoring system. Confrustion was negatively correlated with both immediate and delayed posttest performance across conditions, yet students in the erroneous-example condition experienced more confrustion than problem-solving controls and still outperformed them on the delayed posttest — the condition-level benefit emerged despite, not because of, the elevated confrustion. Notably, Richey et al. treated confusion and frustration as a combined construct. Our data suggest they respond to different antecedents and should be measured separately.

Azevedo et al. [3] studied MetaTutor, an ITS that uses pedagogical agents to scaffold self-regulated learning. More intense initial prompting induced higher confusion in *low* prior knowledge students. MetaTutor's prompts scaffold existing SRL processes, while DeepThought's Plan/Act/Reflect and TVS are domain-specific strategies intended for internalization, not temporary support for skills students already possess. The direction of the ability-confusion interaction may depend on whether the metacognitive support is scaffolding (confusion in those who lack the skill) or instruction (confusion in those who must reconcile new strategies with existing competence).

1.7 Research questions

Three questions test the CVT account, each addressing a different term in Equation 1:

RQ1 How does prior ability moderate which affect, frustration or confusion, responds to metacognitive instruction? (Tests *incongruity*: ability determines whether dominant incongruity is task-level or metacognitive-level.)

RQ2 How does the degree of agency preserved by prompting mode (forced vs. nudge) modulate confusion in high-ability students? (Tests *control*: does agency moderate not only the functional consequences of confusion but also its generation?)

RQ3 How does problem-type choice, an independent agency dimension, further modulate confusion? (Tests a second, independent manipulation of control.)

We additionally test the planning $\times$ pretest interaction on confusion with pretest as a continuous moderator.

2 METHOD

2.1 Participants

Participants were 249 undergraduate Computer Science students in a discrete mathematics course at a large public university. The study was a required lab assignment (Spring 2026). Students were randomly assigned to one of seven experimental conditions ($n = 34$ to 37 per condition).

2.2 Tutoring system

DeepThought is an intelligent tutoring system for teaching propositional logic proof construction [18, 17]. The interface presents the given premises as labeled nodes on a canvas-based workspace and the target conclusion at the bottom. Students iteratively select one or more premise nodes, choose an appropriate logic rule, and apply it to generate a new derived node until they complete the proof. The tutor provides immediate feedback on rule applications and adaptive problem sequencing. The tutor supports forward, backward, and indirect proof construction strategies.

The metacognitive component comprises the two skills: a logic-specific TVS (backward-chaining) and a Plan/Act/Reflect cycle in which students build a plan stack (forward or backward, TVS optional) and execute steps via a "Do it" button, interleaving planning and execution. After each problem, a reflection dialog collects a free-text reflection plus mental-effort (9-point) and help-preference ratings.

Table 1. Study procedure by level.

Level	Problems	Content	Types
1	5	Pretest & introduction	MWE, WE_A, PS, RM
P	3	Planning instruction	WE-P1, WE-P2, PS-PDF
2	4	Plan training begins	PS, PS-PDF, PPE, Buggy
3	4	Plan training continues	PS, PS-PDF, PPE, Buggy
4–6	4 each	Adaptive practice	PS, PPE, Buggy
7	6	Posttest	PS only

2.3 Procedure

The lab averaged approximately 5 hours across 8 levels with 3 to 6 problems each (Table 1).

Level 1 introduced the proof interface through mixed worked examples (MWE), guided worked examples (WE_A), problem-solving (PS), and a rule mastery (RM) exercise. Level P, inserted between Levels 1 and 2, provided planning instruction through two planning worked examples (WE-P1 focuses on plan creation via TVS; WE-P2 focuses on plan execution and parameter filling) followed by a planning dialog problem (PS-PDF), in which the tutor prompts the student through each step of TVS-based planning and then has them execute their own plan. Beginning in Levels 2 and 3, metacognitive conditions diverged: students in planning conditions applied Plan/Act/Reflect and TVS according to their assigned prompting mode. Levels 4 through 6 provided extended practice with adaptive problem sequencing (random or DRL-based, depending on condition). Level 7 served as the posttest, six novel problems solved without metacognitive prompting.

Three problem-type formats appeared from Level 2 onward. *PS* (problem-solving, the primary assessment format) requires the student to construct the entire proof from the given premises to the conclusion. *PPE* (partially worked problem example) presents a subset of the proof's intermediate nodes already on the workspace, and the student fills in the missing derivation steps. *Buggy* presents an erroneous worked example, and the student identifies and corrects faulty steps. The posttest (Level 7) used PS exclusively to assess unprompted performance.

2.4 Experimental conditions

Seven conditions varied along three dimensions (Table 2).

Table 2. Experimental conditions.

tc	Prompting	Selection	Choice	Agency
1	Optional	Random	No	Highest
2	Forced	Random	No	Lowest
3	Nudge	Random	No	Medium
4	Forced	DRL	No	Lowest
5	Nudge	DRL	No	Medium
6	Forced	DRL	Yes	Low
7	Nudge	DRL	Yes	Higher

Prompting mode varied the degree of agency over metacognitive behavior. In *forced* conditions (tc = 2, 4, 6), four designated training problems (2.4, 2.5, 3.4, 3.5) were delivered as guided planning dialogs (PS-PDF) that required a completed plan before the student could proceed; on the remaining problems, planning tools were available but not mandatory. In *nudge* conditions (tc = 3, 5, 7), students received a reminder to plan but could decline and proceed without planning. In the *optional* condition (tc = 1), no planning prompts were presented. Students could use planning tools voluntarily.

Selection strategy determined problem sequencing. In random conditions (tc = 1 to 3), problems within each level were drawn uniformly at random from the level's pool of allowed formats (PS, PPE, Buggy). In DRL conditions (tc = 4 to 7), a deep reinforcement learning policy selected the next problem type based on a 72-feature student model updated after each problem, capturing per-step performance, rule usage, time-on-task, and recent error patterns. From the student's perspective, the two strategies differ in within-level adaptivity, not in the set of formats available.

Problem-type choice was nested within DRL conditions. In choice conditions (tc = 6, 7), before each new problem in Levels 4 through 6, the tutor displayed a selection dialog listing three options (PS, PPE, Buggy); the student chose the format of the upcoming problem. The DRL policy still controlled *which* specific problem instance was delivered for the chosen format, so the student selected the format but not the content. In all other conditions, the DRL policy or random assignment determined the problem type. Problem-type choice was thus an agency dimension orthogonal to prompting mode: a student in a forced prompting condition could still hold problem-type choice, and a student in a nudge condition could still lack it.

2.5 Measures

Prior ability. Pretest performance indexes *domain ability in propositional logic proof construction*, not metacognitive regulation or self-reported confidence. It was computed as the mean problemScore across two Level 1 problem-solving (PS) problems (1.3 and 1.4), administered before any planning instruction (Level P), so the measure is uncontaminated by exposure to TVS or Plan/Act/Reflect. The problemScore composite equally weights rule accuracy, problem completion time, and solution length, normalized by interquartile range within each problem. Pretest scores ranged from 0.31 to 0.93 ($M = 0.693$, $SD = 0.147$, $Mdn = 0.722$). Consistent with the distinction drawn in Section 1.1, high domain ability on this measure does not entail metacognitive proficiency [26].

Posttest. Mean problemScore across six Level 7 problems (7.1 to 7.6), all solved without metacognitive prompting. Problems 7.1 and 7.2 were isomorphic to pretest; 7.3 to 7.6 required transfer to novel proof structures ($M = 0.704$, $SD = 0.116$, $n = 247$).

Affect. Students self-reported their affective state by clicking one of five emoji buttons available throughout the session: Stuck/Frustrated, Confused, Bored, Flow, or Delighted. Reports were voluntary and could be submitted at any time. A total of 7,722 reports were collected from 174 of 249 students (70%), with a mean of 44.4 reports per reporting student ($Mdn = 20.0$). Primary measures were per-student proportions: frustrated_prop (59.5% of all reports) and confused_prop (20.4%). Students with no affect reports were excluded from affect analyses.

Plan quality. For problems with planning, plan quality ($M = 0.875$, $SD = 0.087$) was the ratio of correct plan steps to total steps: planCorrectCount / (planCorrectCount + planErrorCount).

Voluntary metacognitive behavior. Measured on PS problems only (where planning was not mandatory regardless of condition): voluntary planning rate (proportion of PS problems with any plan goal selection; $M = 0.046$, $SD = 0.077$) and TVS adoption rate (proportion of PS problems with any mode toggle to term view; $M = 0.066$, $SD = 0.084$).

2.6 Analysis

Three planned contrasts addressed the research questions: (1) Planning (tc = 2 to 7) vs. Optional (tc = 1), testing whether metacognitive instruction produces different affective profiles; (2) Forced vs. Nudge (tc = 2, 4, 6 vs. tc = 3, 5, 7), testing whether preserved agency modulates confusion; (3) Choice vs. No Choice (tc = 6, 7 vs. tc = 4, 5, within DRL), testing an independent agency dimension. Each contrast was tested with Welch's t-test and Mann-Whitney U, with Cohen's d and 95% CIs. Kruskal-Wallis tests across all seven conditions preceded planned contrasts.

Ability moderation. Each contrast was tested separately within bottom 50% and top 50%, defined by a median split on pretest ($Mdn = 0.722$): $n = 125$ (bottom) and $n = 124$ (top) on the affect sample ($N = 249$, used for Findings 1–3), and $n = 125$ and $n = 122$ on the posttest-complete sample ($n = 247$). To confirm that ability moderation was not a dichotomization artifact, we fit OLS regression with centered pretest as a continuous moderator and tested the planning $\times$ pretest interaction on confused_prop and frustrated_prop.

Exploratory analyses. Within the top 50%, we compared forced and nudge on plan quality, voluntary planning rate, and TVS adoption rate to test whether the same conditions producing more confusion also produced better metacognitive engagement (convergent validity for the effortful learning interpretation). We correlated confused_prop and frustrated_prop with posttest scores, voluntary planning rate, TVS adoption rate, and plan quality within ability groups to test whether confusion was associated with immediate learning outcomes. The ability-split analyses are post-hoc and exploratory. We report them as pattern-generating evidence warranting pre-registered replication, not confirmatory tests.

3 RESULTS

3.1 Full-sample affect: no condition differences

In the full sample ($N = 174$ with affect data), no significant differences in frustrated or confused proportions were observed across the seven conditions (Kruskal-Wallis: frustrated $H = 6.19$, $p = .40$; confused $H = 6.09$, $p = .41$). Frustrated/stuck was the dominant reported affect across all conditions ($M = 0.47$ to 0.62, 59.5% of all reports), while confusion was less frequent ($M = 0.10$ to 0.28, 20.4%). This null provides the baseline against which the ability-moderated findings below should be interpreted.

3.2 Finding 1: ability moderates which affect responds (RQ1)

The full-sample null masks a crossover by prior ability (Table 3).

Table 3. Affect by planning condition and ability group.

Affect	Contrast	Bottom 50%	Top 50%
Frustrated	Plan vs Opt	$d = 0.08$ $p = .79$	$d = -0.68^{*}$ $p = .043$ CI $[-1.33, -0.04]$
Confused	Plan vs Opt	$d = -0.55$ $p = .15$	$d = 0.60$ $p = .24$ (MW) CI $[-0.04, 1.24]$

For top 50% students, metacognitive instruction significantly *reduced* frustration (Planning $M = 0.432$, Optional $M = 0.714$; $d = -0.681$, $p = .043$) and *increased* confusion (Planning $M = 0.217$, Optional $M = 0.029$). The confusion increase is not a shift in the typical student (both medians = 0): the parametric $d = 0.601$, $p < .001$ is inflated by near-zero variance in the small Optional cell ($n = 11$), and the rank test is non-significant (Mann-Whitney $p = .24$). It instead reflects a higher report-level rate of confusion, where high-ability planning students were more likely than Optional students to express confusion on any given report (mixed/GEE logistic OR = 13.4, 95% CI $[2.1, 87.8]$, $p = .007$). This frustration-confusion swap, a robust frustration reduction alongside an elevated confusion rate, is the central finding. Instruction in Plan/Act/Reflect and TVS reduced task-level struggle while introducing metacognitive-level challenge. For bottom 50% students, neither affect differed significantly by condition. Frustration was high across the board and confusion was low, consistent with task-level incongruity dominating the affective experience regardless of metacognitive instruction.

Continuous moderation. Regressing confused_prop on planning condition, centered pretest, and their interaction yielded a significant interaction ($\beta = 1.353$, $t = 2.98$, $p = .003$; $R^2 = .052$ vs. .003 for main effects; $F = 8.89$, $p = .003$). The effect of planning instruction on confusion increases continuously with prior ability. This interaction is robust to estimator (HC3 robust SE $p = .003$; fractional-logit $p = .002$) and does not depend on the median split; we treat it as the primary test for RQ1, with the dichotomized contrast in Table 3 as illustration. A parallel model for frustrated_prop showed a directionally opposite interaction ($\beta = -0.980$, $t = -1.67$, $p = .096$).

Swap by prompting mode. Decomposing the swap by mode shows the two arms behave differently. Relative to Optional, frustration fell under both prompting modes (nudge $d = -0.79$, $p = .02$; forced $d = -0.57$, $p = .09$), and the two modes did not differ from each other on frustration (forced vs. nudge $d = 0.27$, $p = .27$). The confusion increase, by contrast, was concentrated in nudge: forced was indistinguishable from Optional on confusion (Mann-Whitney $p = .71$), and the forced vs. nudge difference was significant (Finding 2). The frustration reduction is thus a general effect of metacognitive instruction, whereas the agency grading that reaches significance is specific to confusion.

3.3 Finding 2: nudge > forced on confusion (RQ2)

Within planning conditions, the forced vs. nudge prompting contrast was tested separately by ability group for both confused and frustrated proportions (Table 4).

Table 4. Confused and frustrated proportions by prompting mode and ability group.

Affect	Ability	Forced M (SD)	Nudge M (SD)	d	p	95% CI
Confused	Top 50%	0.129 (0.250)	0.300 (0.383)	−0.524	.030	[−1.00, −0.05]
Confused	Bottom 50%	0.223 (0.308)	0.189 (0.312)	0.109	.63	[−0.33, 0.55]
Frustrated	Top 50%	0.489 (0.395)	0.378 (0.436)	0.266	.27	[−0.21, 0.74]
Frustrated	Bottom 50%	0.523 (0.424)	0.638 (0.367)	−0.290	.20	[−0.73, 0.15]

Among top 50% students, nudge produced significantly more confusion than forced ($d = -0.524$, $p = .030$). The 95% CI excludes zero. Among bottom 50%, forced and nudge did not differ on confusion ($d = 0.109$, $p = .63$). Because the forced and nudge groups each include a choice condition (tc6 forced, tc7 nudge), this contrast is not independent of problem-type choice; restricted to the no-choice conditions (tc2–5) the effect attenuates but stays directionally consistent ($d = -0.49$, Welch $p = .10$, Mann-Whitney $p = .04$).

Frustration did not reach significance in either ability group, but the directional pattern is informative. In top 50% students, nudge produced numerically *less* frustration than forced ($d = 0.266$, $p = .27$), the directional opposite of the confusion effect: the same prompting mode that elevated confusion also tended to suppress frustration. In bottom 50% students, the direction reversed: nudge produced numerically *more* frustration than forced ($d = -0.290$, $p = .20$), with confusion essentially unchanged. The contrast between ability groups argues against an interpretation that nudge simply increases all negative affect. For high-ability students, preserved agency converts task-level frustration into metacognitive-level confusion. For lower-ability students, who lack the domain resources to capitalize on the freedom, the same preserved agency tends to amplify task-level frustration without engaging the metacognitive challenge that would generate confusion.

This pattern is counterintuitive under a naive reading of the CVT formula: nudge preserves more agency (higher control), which should predict *less* confusion. Droujkov et al. [8] showed that agency moderates the functional relationship between confusion episode dynamics and motivation, where the inverse relationship between average confusion episode duration and intrinsic motivation appeared only under full agency. Our finding extends this: agency moderates not only what confusion *does* but whether it is *generated*. Forced prompting may reduce agency so completely that students comply procedurally without the depth of cognitive engagement required for incongruity to register as confusion. Nudge, by preserving the option to decline, allows identified regulation [25]. Students who choose to engage with the strategy process deeply enough for the incongruity to produce confusion rather than mere procedural compliance.

Supporting evidence from metacognitive behavior. Within the top 50%, nudge produced marginally higher plan quality ($d = -0.358$, $p = .068$), while voluntary planning rate ($d = 0.276$, $p = .16$) and TVS adoption ($d = 0.064$, $p = .74$) did not differ. In the bottom 50%, the plan quality advantage of nudge over forced was large ($d = -0.993$, $p < .0001$). Because forced prompting mandated a completed plan on the designated planning-dialog problems while nudge did not, forced students produced more plan steps on those problems; the nudge ratio advantage therefore reflects more selective, higher-accuracy engagement among those who choose to plan rather than a greater volume of correct metacognitive practice.

3.4 Finding 3: choice trends toward more confusion (RQ3)

A second, independent agency dimension, problem-type choice (tc = 6, 7 vs. tc = 4, 5, within DRL), showed a parallel pattern in top 50% students (Table 5).

Table 5. Affect by choice availability (top 50% only).

Affect	Choice M (SD)	No Choice M (SD)	d	p	95% CI
Confused	0.327 (0.391)	0.151 (0.289)	0.524	.099	[−0.06, 1.11]
Frustrated	0.440 (0.427)	0.437 (0.421)	0.005	.99	[−0.57, 0.58]

The effect on confusion was in the same direction and of the same magnitude as Finding 2 ($d = 0.524$), but marginal ($p = .099$). The 95% CI includes zero. Frustration was entirely unaffected by choice ($d = 0.005$, $p = .99$). This result is consistent with the CVT account, where a second, independent manipulation of perceived control produces the same directional effect on confusion without affecting frustration, but it does not reach conventional significance with the available sample size ($n = 20$ choice, $n = 27$ no choice within the top 50%).

Choice conditions also reported significantly fewer total affect events ($M = 19.2$ vs. 33.4, Mann-Whitney $p = .009$, $d = -0.33$), suggesting problem-type choice provides an agency outlet reducing the overall impulse to report affect. When students in choice conditions did report, a greater proportion of their reports were confusion rather than frustration.

3.5 Confusion and learning outcomes

Within the top 50%, confusion was unrelated to overall posttest (Spearman $\rho = -0.011$, $p = .92$, $n = 79$), isomorphic posttest ($\rho = -0.009$, $p = .94$), and transfer posttest ($\rho = -0.023$, $p = .84$). Frustration showed a weak negative association with isomorphic posttest ($\rho = -0.226$, $p = .045$) but not overall or transfer posttest. In the bottom 50%, neither affect correlated with any posttest measure (all $|\rho| < 0.17$, all $p > .11$).

Confusion was also unrelated to metacognitive behavior within the top 50%: confused_prop did not correlate with voluntary planning ($\rho = 0.145$, $p = .20$), TVS adoption ($\rho = 0.079$, $p = .49$), or plan quality ($\rho = 0.044$, $p = .70$). The confusion observed under metacognitive instruction is neither productive (associated with better learning) nor harmful (associated with worse learning) within a single session. It is orthogonal to immediate performance outcomes.

4 DISCUSSION

4.1 A unified CVT account

The three findings converge on a single mechanism within CVT's framework (Equation 1).

Finding 1 addresses the *incongruity* term. For high-ability students, Plan/Act/Reflect and TVS are genuinely new metacognitive skills that create strategy-level incongruity. The instructed approach extends beyond or conflicts with their existing problem-solving methods. These students have sufficient domain competence to handle the proof tasks themselves (frustration drops), but they experience confusion from the unfamiliar metacognitive process. For lower-ability students, the dominant incongruity is at the task level. The proofs themselves are challenging, producing frustration rather than confusion.

Finding 2 addresses the *control* term, but not in the direction a naive CVT reading predicts. Forced prompting reduces control more than nudge, yet produces *less* confusion. Droujkov et al. [8] showed that agency moderates the functional consequences of confusion, where the confusion-motivation link only appeared under full agency. Our data go further, showing that agency also moderates the *generation* of confusion. Forced prompting produces behavioral compliance without the cognitive engagement needed for incongruity to be appraised. Nudge preserves enough agency for identified regulation [25]. Students who choose to engage with the strategy process it deeply enough for the incongruity to register. Control operates not as a simple denominator but as a threshold: below a minimum level of agency, genuine engagement does not occur, and confusion is suppressed.

Finding 3 provides a partial replication with an independent manipulation of control. Problem-type choice, a second agency dimension, produces the same directional effect ($d = 0.524$) on confusion in high-ability students, though the effect is marginal. The convergence across two independent manipulations of agency, prompting mode (Finding 2) and problem-type choice (Finding 3), strengthens the inference that control is the operative CVT parameter, even though neither manipulation alone would be definitive.

4.2 Confusion as effortful metacognitive learning

The frustration-confusion swap in high-ability students (Finding 1) is the primary evidence favoring effortful learning over scaffolding mismatch. Three observations support this interpretation. First, frustration drops ($d = -0.681$) while confusion rises ($d = 0.601$). Pure mismatch predicts both negative affects increase or frustration remains flat. Second, nudge produces more confusion than forced ($d = -0.524$). If confusion were a reaction to unnecessary constraint, the more controlling condition should produce more. Third, nudge produces higher plan-quality ratios than forced (overall $d = 0.66$, bottom 50% $d = 0.993$; nudge higher in both). The conditions generating more confusion also generate higher-accuracy (though lower-volume) planning.

The two sources of incongruity, scaffolding mismatch and effortful learning, are not mutually exclusive, and the current data cannot definitively separate them. However, the overall pattern is more consistent with productive metacognitive challenge. D'Mello and Graesser [9] describe exactly this kind of productive confusion: cognitive disequilibrium at an impasse that, if resolved, drives deeper processing. Lodge and Kennedy's [16] model of optimal vs. suboptimal confusion zones provides further theoretical grounding. The frustration-confusion swap suggests high-ability students are in the optimal zone (productively challenged by new metacognitive skills), not the sub-optimal zone (overwhelmed by unnecessary constraint).

Plan/Act/Reflect and TVS create productive impasses because they are domain-specific skills requiring new learning, not generic "plan ahead" prompts (see Section 2.2). Even students who performed well on the domain tasks had no prior experience with these structured approaches to logic proof construction. As Schoenfeld [26] demonstrated, expert problem solvers employ metacognitive control processes that novices lack, and the control process itself is cognitively demanding even when the domain content is not.

The null confusion-posttest relationship (Section 3.5) neither supports nor undermines this interpretation. Richey et al. [23] observed a similar dissociation: confrustion correlated negatively with posttest performance, yet the erroneous-example condition, despite more confrustion than controls, outperformed them on the delayed posttest. Within a single 5-hour session, metacognitive skills may be too nascent to manifest as measurable performance gains. Students are still learning to plan and apply TVS, not yet reaping benefits from internalized strategies. This temporal dissociation between metacognitive learning and domain performance is expected when two skill dimensions are developing on different timescales.

4.3 Agency as moderator of confusion dynamics

Droujkov et al. [8] showed that agency moderates the *functional consequences* of confusion: the inverse relationship between the average duration of the confusion episode and intrinsic motivation appeared only under full agency. Our data extend this by showing that agency also moderates the *generation* of confusion in high-ability students ($d = -0.524$, $p = .030$ for prompting; $d = 0.524$, $p = .099$ for choice). Together, the two studies suggest that agency operates at multiple levels of the confusion process and that the relationship is ability-moderated: only high-ability students show the confusion response to agency manipulations, while lower-ability students' affective responses are dominated by task-level challenge regardless of agency.

4.4 Design implications

These findings have four implications for the design of affect-aware tutoring systems.

First, confusion under metacognitive instruction is diagnostic rather than uniformly negative. In high-ability students, it signals engagement with new metacognitive skills, and removing instruction in response would terminate the productive cognitive disequilibrium the system was designed to support. Lodge and Kennedy's [16] zones model suggests the appropriate response is to maintain the learner in this optimal confusion zone rather than eliminating confusion entirely.

Second, frustration and confusion have distinct antecedents and impacts and should be measured with sufficient resolution to recover that distinction [4], rather than collapsed into a single "confrustion" construct [23]. Frustration in lower-ability students signals task-level struggle; persistence support, hints, or reduced problem difficulty are appropriate. Confusion in higher-ability students signals metacognitive-level challenge; continued engagement with the strategy may be sufficient.

Third, prompting should preserve agency. Nudge produces both higher confusion and higher-accuracy planning than forced prompting. The agency preserved by nudge enables the genuine cognitive engagement that drives both productive confusion and more accurate planning among those who plan. Forced prompting achieves higher-volume but lower-accuracy planning on the mandated problems, with voluntary uptake comparable to nudge but less of the confusion that marks deeper processing.

Fourth, affect interpretation requires ability context, and the direction of the ability-confusion interaction depends on whether metacognitive support is scaffolding or instruction. Lehman et al. [15] found that only students with high cognitive ability and drive benefit from induced confusion, and Azevedo et al. [3] found that MetaTutor's SRL prompts, which scaffold existing self-regulation processes, induced more confusion in low prior knowledge students, while our domain-specific metacognitive instruction induced more confusion in high-ability students. Systems should distinguish between scaffolding existing skills (where confusion signals insufficient support) and teaching new skills (where confusion signals productive engagement), and should not act on a detected affect without also conditioning on ability.

5 LIMITATIONS AND FUTURE DIRECTIONS

The affect measure relies on voluntary emoji self-reports, which depend on students' willingness and ability to introspect on their affective states. Of 249 students, 174 (70%) reported any affect, and proportions are sensitive to reporting frequency. Behavioral or physiological measures of confusion would provide convergent validity.

The ability-moderation analyses are exploratory. The median split on pretest is post-hoc and not pre-registered. After splitting by both ability and condition, sample sizes reduce to $n = 15$ to 20, insufficient for confirmatory inference. The continuous moderation ($p = .003$) mitigates the dichotomization concern but does not address the post-hoc nature.

Finding 3 (choice effect) is marginal ($p = .099$) and should be interpreted cautiously. Problem-type choice is confounded with DRL-based problem selection. Choice was only available under DRL conditions (tc = 6, 7), so the choice effect cannot be fully isolated from the effects of the problem selection strategy. The forced vs. nudge contrast (Finding 2) is likewise not fully independent of choice, since the choice conditions sit within it; removing them attenuates the effect to $p = .10$ (Welch), though the nonparametric test remains significant ($p = .04$).

The CVT formula is used as a conceptual framework, not a quantitative model. The parameters (λ, incongruity, value, control) are not measured directly. We infer incongruity from the novelty of the metacognitive instruction, and control from the experimental manipulation of agency. Direct measurement of perceived control and perceived incongruity would allow more precise tests of the appraisal mechanism.

The effortful learning and scaffolding mismatch interpretations cannot be definitively distinguished. Think-aloud protocols or process-level confusion tracking could reveal whether confusion diminishes as metacognitive strategies become familiar (consistent with effortful learning) or persists throughout the session (more consistent with mismatch). Level-by-level affect data from this study show suggestive but underpowered trends: confusion in top 50% planning students peaks at training levels and decreases toward the posttest, but per-level cell sizes are too small for reliable inference.

Future work should pursue four directions. First, a pre-registered ability-stratified design with adequate per-cell power ($n = 30$+ per ability group $\times$ condition cell) to confirm the aptitude-treatment interaction on confusion [15]. Second, longitudinal tracking across multiple sessions, motivated by Richey et al.'s [23] finding that confusion predicted delayed but not immediate posttest gains. Repeated sessions would test a central prediction of the productive-confusion hypothesis: confusion should diminish as students grow familiar with Plan/Act/Reflect and TVS, whereas confusion that persists across sessions would instead indicate a lasting scaffolding mismatch. Third, direct manipulation of CVT components to test the appraisal mechanism rather than inferring it from conditions. Fourth, a within-study comparison of scaffolding vs. instruction conditions to test whether the ability-confusion interaction reverses.

6 CONCLUSION

Confusion under metacognitive instruction in an intelligent tutoring system is not a uniform signal of difficulty. It is an ability-moderated, agency-dependent response that, in high-ability students, co-occurs with reduced frustration and more accurate planning when students choose to plan. Control-Value Theory provides a coherent framework: the source of incongruity (task-level vs. metacognitive-level) determines which affect responds, while the degree of agency moderates both the generation and the functional consequences of confusion. Although theoretical work has recognized confusion's productive potential [10, 16, 14], affect-aware system implementations typically treat detected confusion as a negative state to be remediated through hints or difficulty reduction [7, 12]. Our findings suggest this default response is inappropriate when confusion arises from metacognitive instruction in high-ability students, and that prompting designs preserving learner agency, nudge rather than forced, produce both richer affective engagement and better metacognitive outcomes.

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